

Anish Saxena

PHD CANDIDATE · SCHOOL OF COMPUTER SCIENCE · GEORGIA INSTITUTE OF TECHNOLOGY

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Research Summary

My research interests include efficient AI, datacenter systems, and hardware security, with a current focus on architecting hardware and software to scale the throughput and interactivity of LLM serving systems. My work is guided by workload-aware performance models and simulations, co-designed across the stack, and validated through system-building.

Education

Georgia Institute of Technology

Atlanta, USA

DOCTOR OF PHILOSOPHY, COMPUTER SCIENCE · GPA: 3.94/4.00

2021 - 2026 (expected)

- Advisor: Prof. Moinuddin K. Qureshi
- Research: Efficient AI, Datacenter Systems, and Hardware Security
- Minor in AI Strategy and Economics

Indian Institute of Technology Kanpur

Kanpur, India

BACHELOR OF TECHNOLOGY, MECHANICAL ENGINEERING · CPI: 9.1/10.0

2017 - 2021

- Advisor: Prof. Biswabandan Panda
- Minor in Computer Systems

Honors

2025	NVIDIA Graduate Fellowship , Among 10 students worldwide, 600 applicants	USA
2025	GT College of Computing Outstanding GRA Award , Sole recipient, 450 candidates	Atlanta
2021	EPFL EDIC Fellowship (Declined) , Awarded by EPFL and Swiss Government	Lausanne
2019	Semiconductor Research Corporation (SRC) Member , Sole undergrad from India	India
2017	Aditya Birla Scholarship , Among 15 students from IIT and BITS, 160 applicants	Mumbai
2017	All India Rank 1828 , Joint Entrance Examination Advanced, 175,000 students	India
2017	KVPY Fellowship (Declined) , Awarded by IISc Bangalore and Government of India	Bangalore

Publications

CONFERENCE PAPERS

Citadel: Rethinking Memory Allocation to Prevent Inter-Domain Rowhammer Exploits

Anish Saxena, Walter Wang, Alexandros Daglis

58th IEEE/ACM International Symposium on Microarchitecture (MICRO), Seoul, South Korea, October 2025.

CXL-Centric Memory System for Scalable Servers

Albert Cho*, Anish Saxena*, Moinuddin Qureshi, Alexandros Daglis

*Equal contribution

36th International Conference for High Performance Computing, Networking, Storage, and Analysis (SC), Atlanta, USA, November 2024.

ImPress: Securing DRAM Against Data-Disturbance Errors via Implicit Row-Press Mitigation

Anish Saxena, Aamer Jaleel, Moinuddin Qureshi

57th IEEE/ACM International Symposium on Microarchitecture (MICRO), Austin, USA, November 2024.

Rubix: Low Overhead Secure Rowhammer Mitigations via Randomized Memory Mapping

Anish Saxena, Saurav Mathur, Moinuddin Qureshi

29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), San Diego, USA, April 2024.

START: Scalable Tracking for Any Rowhammer Threshold

Anish Saxena and Moinuddin Qureshi

30th IEEE International Symposium on High-Performance Computer Architecture (**HPCA**), Edinburgh, Scotland, March 2024.

PT-Guard: Integrity-Protected Page Tables against Breakthrough Rowhammer Attacks

Anish Saxena, Gururaj Saileshwar, Jonas Juffinger, Andreas Kogler, Daniel Gruss, Moinuddin Qureshi

53rd IEEE/IFIP Conference on Dependable Systems and Networks (**DSN**), Porto, Portugal, June 2023.

AQUA: Scalable Rowhammer Mitigation by Quarantining Aggressor Rows at Runtime

Anish Saxena, Gururaj Saileshwar, Prashant Nair, Moinuddin Qureshi

55th IEEE/ACM International Symposium on Microarchitecture (**MICRO**), Chicago, USA, October 2022.

JOURNAL PAPERS

Scalable Multi-node Fast Fourier Transform on GPUs

Manthan Verma, Soumyadeep Chatterjee, Gaurav Garg, Bharatkumar Sharma, Nishant Arya, Shashi Kumar, Anish Saxena, Mahendra K. Verma

4th Edition of **SN Computer Science**, 625, Springer Nature Singapore, 2023.

WORKSHOP PAPERS AND POSTERS

Cascade: Utility-Driven Speculative Decoding for Mixture-of-Experts

Anish Saxena, Po-An Tsai, Hritvik Taneja, Aamer Jaleel, Moinuddin Qureshi

9th Young Professionals Symposium (**YPS**), co-located with MLSys, Seattle, USA, May 2026.

DABANGG: A Case for Noise Resilient Flush-Based Cache Attacks

Anish Saxena and Biswabandan Panda

16th IEEE Workshop on Offensive Technologies (**WOOT**), co-located with S&P, San Francisco, USA, May 2022.

UNDER REVIEW

CARMA: Concurrent Access to Remote Memory and HBM to Accelerate LLM Inference

Anish Saxena, Jae Hyung Ju, Hritvik Taneja, Moinuddin Qureshi, Po-An Tsai, Aamer Jaleel, Christos Kozyrakis

Midas: Utility-Driven Speculative Decoding for Mixture-of-Experts

Anish Saxena, Po-An Tsai, Hritvik Taneja, Aamer Jaleel, Moinuddin Qureshi

SiFAR: Synchronization-Free All-Reduce for Low-Latency LLM Inference

Hritvik Taneja, Anish Saxena, Jae Hyung Ju, Neal Crago, Moinuddin Qureshi

TileLens: Efficiently Using Large-Granularity Memory Systems with Transparent Two-Dimensional Memory Layout

Jae Hyung Ju, Euijun Chung, Hritvik Taneja, Anish Saxena, Shinnung Jeong, Hyesoon Kim, Moinuddin Qureshi

SplitStream: Maximizing System Bandwidth Utilization via Read-Write Buffer Management

Hritvik Taneja, Jae Hyung Ju, Anish Saxena, Moinuddin Qureshi

Patents

Hybrid HBM-High-Bandwidth Interconnects Base Layer

Bill Dally, Anish Saxena, Aamer Jaleel, Po-An Tsai, Christos Kozyrakis

Attorney Docket Number 6700.092US1, **Filed** October 2025 to US Patent and Trademark Office.

Data Placement on Multi-Tier GPU Memory

Anish Saxena, Aamer Jaleel, Po-An Tsai, Christos Kozyrakis

Attorney Docket Number 6700.089PRV, **Filed** October 2025 to US Patent and Trademark Office.

Processor Memory Scaling via High-Bandwidth Interconnects

Anish Saxena, Aamer Jaleel, Po-An Tsai, Christos Kozyrakis

Attorney Docket Number 6700.091US1, **Filed** October 2025 to US Patent and Trademark Office.

Improving Mixture-of-Expert Large Language Models by Using Restricted Set of Experts

Anish Saxena, Po-An Tsai, Bingyao Li, Aamer Jaleel

Attorney Docket Number 515137/24-WE-0980US01, **Filed** January 2025 to US Patent and Trademark Office.

Dynamic Key Value Pair Cache Scheduling

Bingyao Li, Aamer Jaleel, Po-An Tsai, **Anish Saxena**

Attorney Docket Number 24-BO-0923-US02, **Filed** January 2025 to US Patent and Trademark Office.

Experience

Future Architectures & Systems Lab, Georgia Tech

Prof. Moinuddin K. Qureshi

GRADUATE RESEARCH ASSISTANT

Aug. 2021 - present

- Designing and architecting computer systems to tackle challenges in LLM serving, datacenters, and security.
- Published top-tier first-author papers (MICRO, HPCA, ASPLOS, SC) and delivered seminars and invited talks.

NVIDIA Research, USA

Dr. Aamer Jaleel

RESEARCH INTERN, SYSTEMS ARCHITECTURE RESEARCH GROUP

May. 2025 - Aug. 2025

- Developed Atlas, a slack timing based performance model, for rack-scale systems serving recent LLMs.
- Designed BOOST, a data access manager to maximize system bandwidth and improve throughput by $2.4\times$.
- Architected NVLink-Attached Memory (NAM) at NVSwitch and at SoC to scale throughput by $2.7\times$ to $7.3\times$.

NVIDIA Research, USA

Dr. Po-An Tsai

RESEARCH INTERN, ARCHITECTURE RESEARCH GROUP

May. 2024 - Aug. 2024

- Developed a roofline model to analyze serving of Mixture-of-Expert Large Language Models (MoE-LLMs).
- Improved throughput of MoE-LLMs by $3\times$ in long context serving scenarios by restricting set of usable experts.

AMD Research, USA

Dr. Shaizeen Aga

RESEARCH INTERN, APPLICATIONS-DRIVEN ARCHITECTURE TEAM

May. 2023 - Aug. 2023

- Developed a roofline model to analyze training of Large Language Models (LLM) on thousands of GPUs.
- Designed a DDR-centric GPU design with better LLM training scalability compared to HBM-centric designs.

Micron Technology, USA

Patrick Estep

RESEARCH INTERN, ADVANCED MEMORY GROUP

May. 2022 - Aug. 2022

- Architected CXL-enabled memory systems to accelerate performance of Google's datacenter workloads.
- Developed a multi-core emulation framework to replay datacenter workload traces on native hardware.

NVIDIA Corporation, India

Bharatkumar Sharma

HPC GPU ADVOCATE INTERN, OPENACC AND GPU HACKATHONS TEAM

May 2021 - Aug. 2021

- Created multi-node GPU profiling-driven programming guides for HPC with topology-aware communication.
- The open-source tutorials and code are accessible at github.com/openhackathons-org/nways_multi_gpu/.

Intel Labs, India

Anant Nori

RESEARCH INTERN, PROCESSOR ARCHITECTURE RESEARCH LAB

May 2020 - Sep. 2020

- Improved performance of non-inclusive caches by extending reuse-distance-based cache policies.
- Accelerated in-house cycle-level simulator and collected memory traces to perform functional simulations.

CAR3S Group, IIT Kanpur

Prof. Biswabandan Panda

GROUP MEMBER AND SRC STUDENT MEMBER

Apr. 2019 - Jun. 2020

- Devised DABANGG (WOOT'22), refinements that enable accurate and noise-resilient cache attacks.
- Developed tracing techniques for ARM architectures and analyzed cache compression for mobile SoCs.

New York Office, IIT Kanpur

COMPUTER SYSTEMS INTERN

Prof. Manindra Agrawal

May 2018 - Jul. 2018

- Led a team of 4 to develop the infrastructure stack of a multi-node microservices-based Kubernetes cluster.
- Configured Spinnaker-based CI/ CD pipeline and integrated canary analysis and stress testing capabilities.

Mentorship

Rya Sanovar

PHD STUDENT

Georgia Tech

Aug. 2025 - present

- Developing techniques to improve RAG-based LLM inference systems through sparse KV computation.

Jae Hyung Ju

PHD STUDENT

Georgia Tech

Jan. 2025 - present

- Designing runtime systems to concurrently access remote memory and HBM to accelerate LLM inference.

Hritvik Taneja

PHD STUDENT

Georgia Tech

Jan. 2024 - present

- Developing software techniques to overlap barriers with data transfer for sync-free all-reduce in LLMs.

Walter Wang

PHD STUDENT

Georgia Tech

Aug. 2023 - Oct. 2025

- Developed a new memory allocator in Linux that is resilient to inter-domain Rowhammer exploits.

Caroline Huang

PHD STUDENT

Georgia Tech

Aug. 2023 - Nov. 2024

- Designed adaptive Refresh Management (RFM) scheduling policies that scale to low Rowhammer threshold.

Saurav Mathur

PHD STUDENT

Georgia Tech

Jan. 2023 - Apr. 2023

- Developed randomized memory mapping to minimize hot-rows, enabling scalable Rowhammer mitigations.

Sibi Renganath Sudhakar

MS STUDENT → ASIC LOW POWER DESIGNER AT NVIDIA

Georgia Tech

Jan. 2023 - Apr. 2023

- Analyzed memory reservation primitives to enable isolation-based protection against Rowhammer attacks.

Service

2026	IEEE/ ACM MICRO Conference, Reviewer	GaTech
2026	IEEE Computer Architecture Letters, Reviewer	GaTech
2025	IEEE Transactions on Big Data, Reviewer	GaTech
2025	IEEE Micro, Reviewer	GaTech
2025	IEEE HPCA 2026 Conference, Reviewer	GaTech
2025	IEEE Computer Architecture Letters, Reviewer	GaTech
2025	IEEE Transactions on Computers, Reviewer	GaTech
2024	IEEE Computer Architecture Letters, Reviewer	GaTech
2024	SCS Graduate Student Association, Co-Chair of Events, Co-Chair of Faculty Affairs	GaTech
2023	Secure and Reliable Computer Architecture, Head TA	GaTech
2023	SGA Graduate Student Wellness Committee, Lead, Bill of Rights Team	GaTech
2023	IEEE HPCA 2024 Conference, Artifact Evaluation Committee	Edinburgh
2023	Secure and Reliable Computer Architecture, Head TA	GaTech
2022	Introduction to Quantum Computing, Course Development TA	GaTech
2020	Systems Reading Group, Leader	IIT Kanpur
2019	Programming Club, Coordinator	IIT Kanpur

Talks

2026	Cascade: Utility-driven Speculation for MoEs , poster presentation at YPS-MLSys	Seattle
2026	Breaking the GPU Memory Wall , Graduate Fellowship talk at Nvidia GTC 2026	San Jose
2025	Citadel: RH-aware Memory Allocation , paper talk at 58 th MICRO conference	Seoul
2025	Rethinking Systems for Serving Reasoning LLMs , end-of-intern talk at Nvidia	Santa Clara
2024	CoaXiaL: CXL-centric Server Systems , paper talk at 36 th SC conference	Atlanta
2024	ImPress: Implicit RowPress Mitigation , paper talk at 57 th MICRO conference	Austin
2024	High-Throughput Mixture-of-Expert LLM Serving , end-of-intern talk at Nvidia	Westford
2024	Rubix: Randomized Memory Mappings , paper talk at 29 th ASPLOS conference	San Diego
2024	START: Scalable Rowhammer Tracking , paper talk at 30 th HPCA conference	Edinburgh
2024	Scalable Rowhammer Protection at Ultra-low Cost , invited guest talk	IIT Bombay
2023	Memory System Design for Scalable LLM Training , end-of-intern talk at AMD	Santa Clara
2023	PT-Guard: Integrity-Protected Page Tables , paper talk at 53 rd DSN conference	Portugal
2023	Secure and Scalable Rowhammer Defenses , invited guest talk at SAFARI Group	ETH Zurich
2022	AQUA Rowhammer Mitigation , paper talk at the 55 th MICRO conference	Chicago
2022	Rowhammer Attacks and Defenses , invited guest talk at CASPER Group	IIT Bombay
2022	Analyzing Google Datacenter Workload Traces , invited talk at Intel Labs	Bangalore
2022	Replaying Google Datacenter Workload Traces , end-of-intern talk at Micron	Dallas
2022	DABANGG Attack , paper talk at the 16 th WOOT workshop	San Francisco
2021	CUDA Programming , guest lecture, course on High Performance Computing	IIT Kanpur
2020	Noise-resilient Flush Attacks , CAOS reading group to students and faculty	IIT Kanpur
2020	Microarchitectural Security , talk and demo at SRC Annual Design Review	Bangalore
2019	Flush-based Attacks , guest lecture, course on Secure Memory Systems	IIT Kanpur
2019	ZombieLoad and CLKScrew Attacks , CAOS reading group to students and faculty	IIT Kanpur

Coursework

• Advanced OS: Distributed Systems ^A	• Machine Learning ^A	• Secure & Reliable CompArch ^A
• Experimental Research in IS ^A	• Economics of AI & ML ^A	• Operations Strategy ^A
• High Performance CompArch ^A	• Parallel CompArch ^A	• Computer Architecture ^{A*}
• High Performance Computing ^{A*}	• High Performance Programming ^A	• Modern Cryptology ^A
• Topics in OS: Programming NVM	• Operating Systems ^A	• Quantum Computing ^A
• Computer Organization ^A	• Data Structures & Algorithms	• Non Classical Logic
• Introduction to Economics ^{A*}	• Financial Markets ^O	• Linear Algebra
<i>A*: grade for exceptional performance</i>	<i>A: grade O: online course</i>	

Miscellaneous

• Mentored MS and PhD students on technical projects and advised them on navigating grad school.	2021-present
• Gave an invited talk, as a "pioneering alumnus" of IIT Kanpur, on research career opportunities in CS.	2023
• Mentored 15 students at IIT Kanpur; helped them navigate academic and career challenges in college.	2020
• Represented CAR3S group in departmental seminars and maintained the group's digital presence.	2019, 2020